



MODULE – 2

PROTOCOLS IN IOT INFRASTRUCTURE

MESSAGING PROTOCOLS

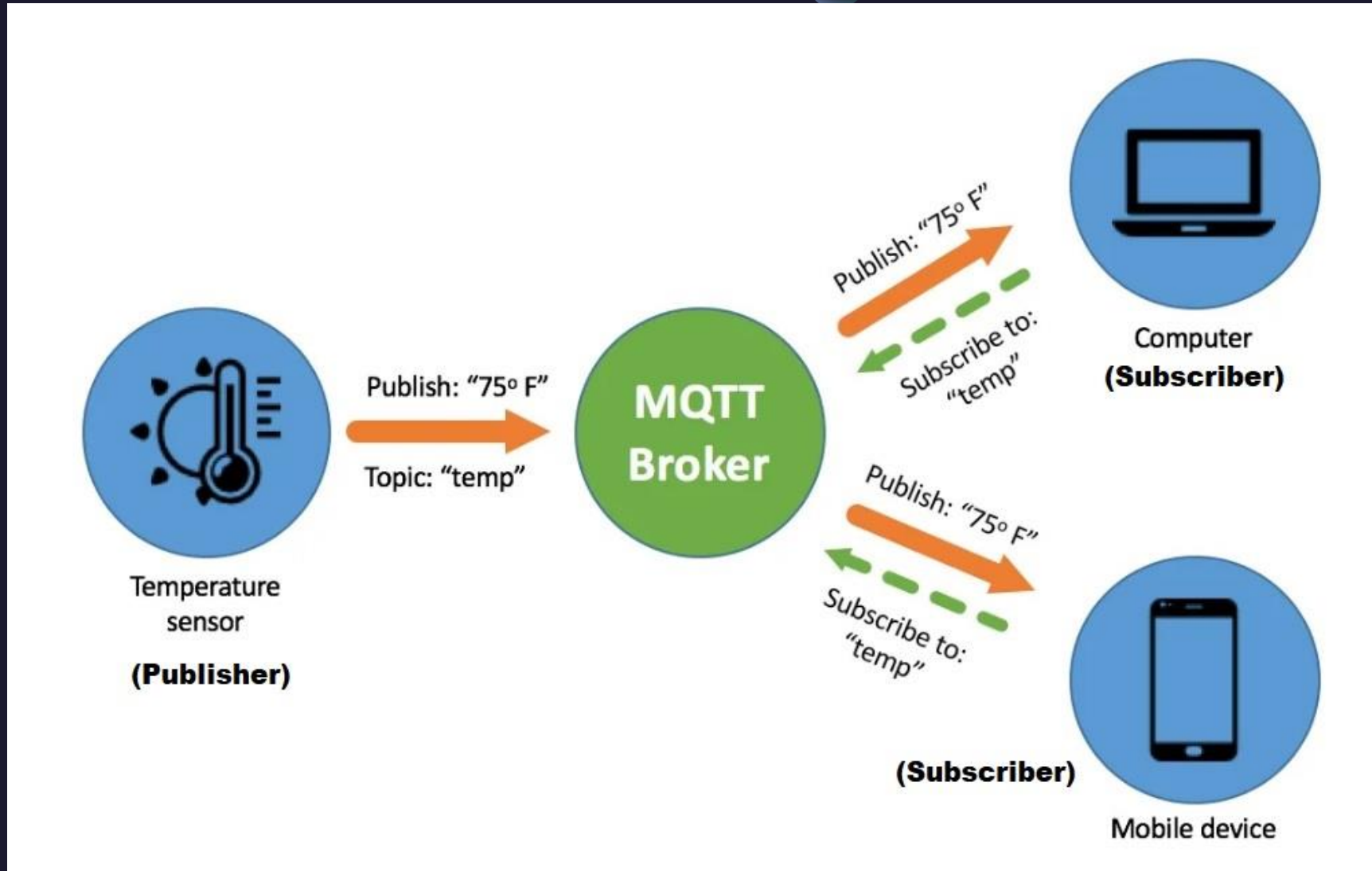
Messaging protocols are used to sent data/message to and from the cloud for IoT application.

- The two important messaging protocols are:
 1. MQTT – Message Queuing Telemetry Transport
 2. CoAP – Constrained Application Protocol

MQTT – Message Queuing Telemetry Transport

- It is a protocol used to send or receive data to and from the cloud in IoT applications.
- It is a **light weight** protocol.
- It uses **Publish – Subscribe** method.

MQTT – Message Queuing Telemetry Transport





MQTT components are:

1. **Message** : This is the data to be send or received.
 2. **Publisher** : Collect the data and publish to the broker.
 3. **Topic** : Data is labelled as topics
 4. **Broker** : Send messages to the subscribers of that topic.
 5. **Subscriber** : Subscribe a topic and receive message from broker.
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MQTT Working

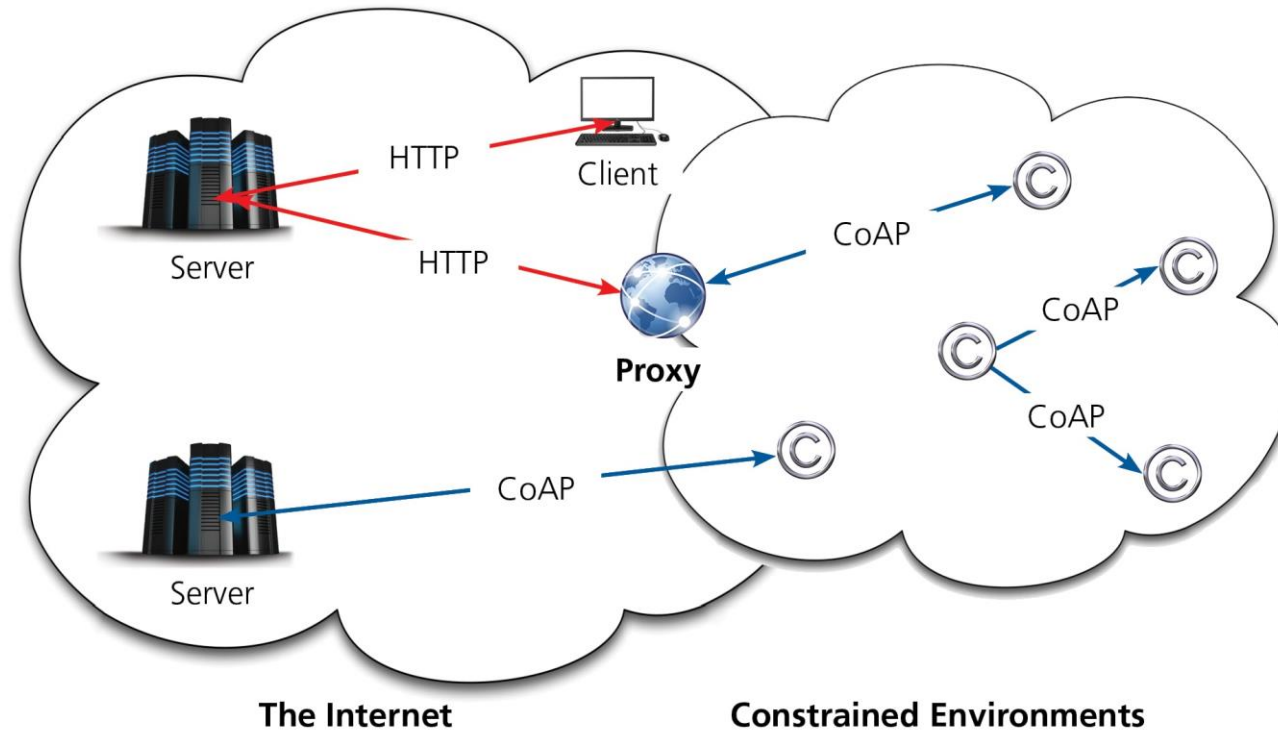


- Publisher collect the data and publish as message to the MQTT broker.
 - The message is labelled as topics.
 - The MQTT broker will forward the message to the subscribers.
 - MQTT broker may be server or cloud.
 - The subscriber can also be a publisher.
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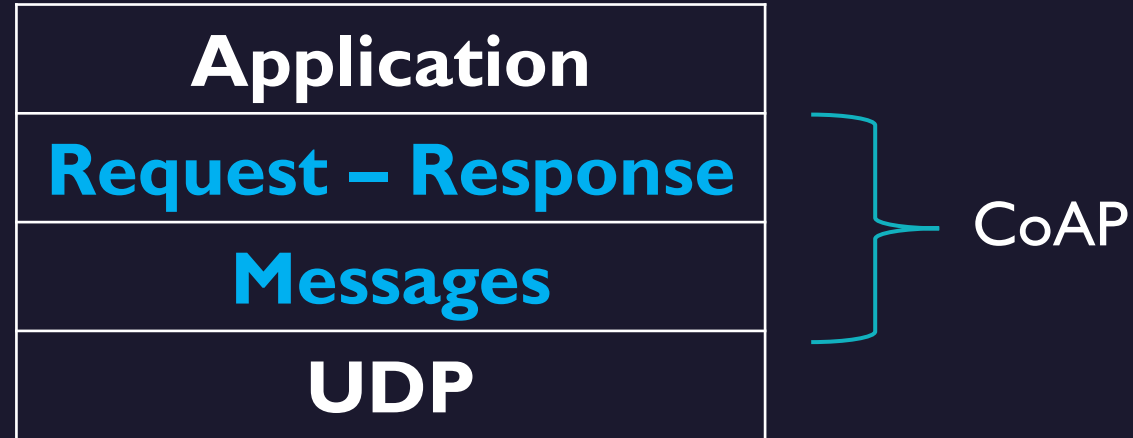
CoAP – Constrained Application Protocol

- It is a special protocol designed to work with constrained hardware
- It is similar to HTTP
- It is designed for IoT networks
- It is based on Client – Server architecture
- It is a one-to-one communication protocol
- It uses GET/PUT/DELETE methods.

CoAP Architecture



Layers of CoAP

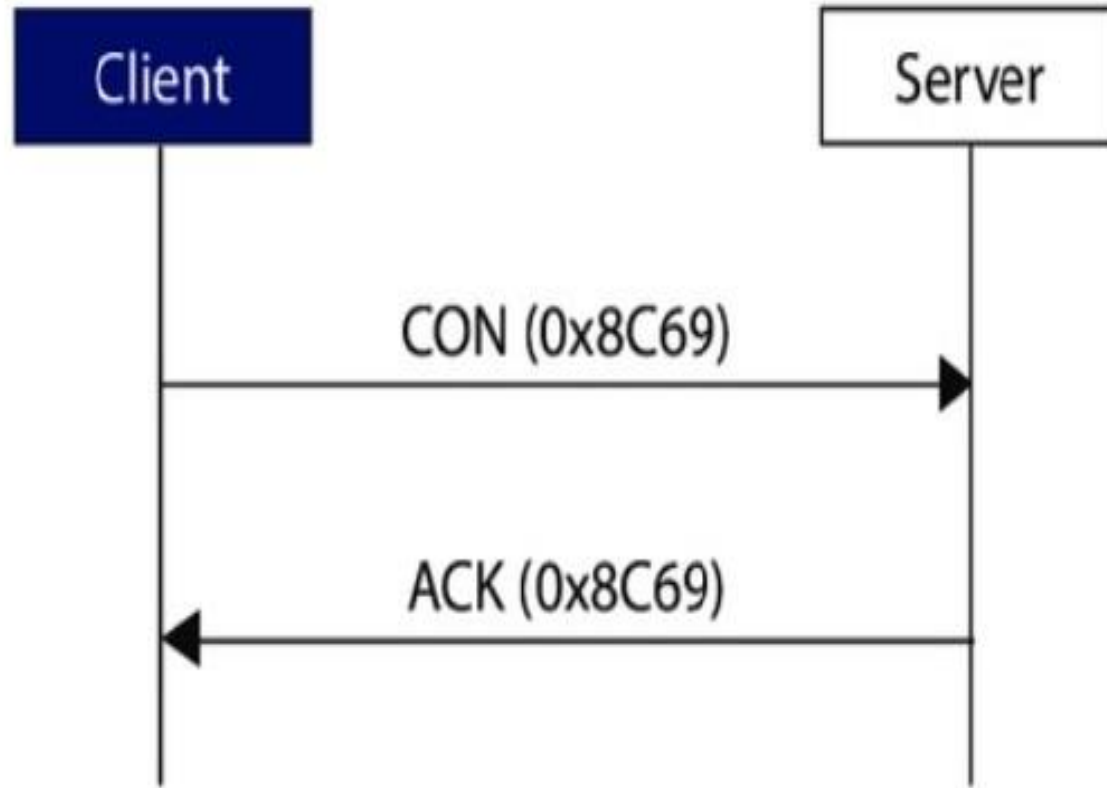


- CoAP is a two layer protocol
- The lower layer is the message layer
- The upper layer is called Request – Response layer
- It works on top of UDP

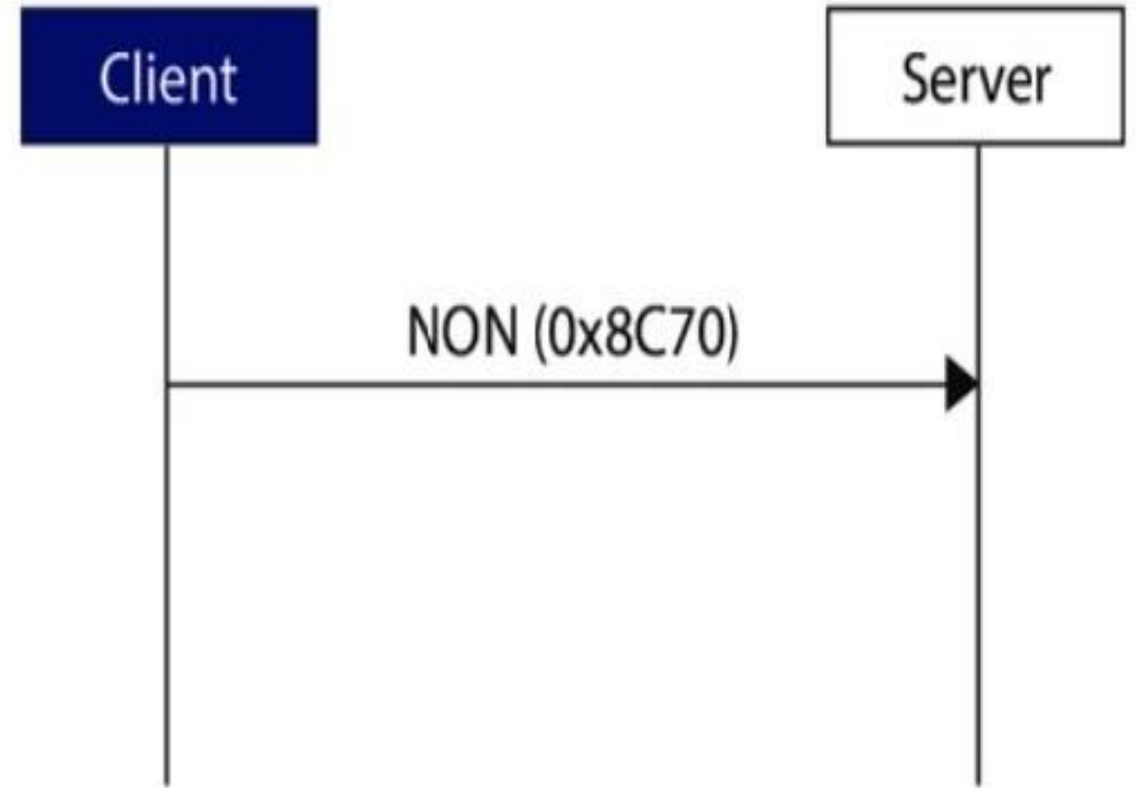
Message Layer

CoAP supports four types of messages:

1. Confirmable (**CON**) – In this type, an acknowledgment should be send after receiving the message. When there is a timeout, a reset (RST) message will be sent.
2. Non-Confirmable(**NON**)- In this type an ID will be send as part of the message. No acknowledgement is send by the receiver.
3. Acknowledgement (**ACK**) – Used to send acknowledgment
4. Reset (**RST**) – Used to initiate reset when time out occur.



(a) Confirmable Message



(b) Non-confirmable Message

Examples of Confirmable and Non-confirmable CoAP messages.

Request-Response Layer

1. Piggy-Backed:

In this method, client send request through CON message and its response is send through ACK message.

2. Separate Response:

When a CON message is send to the server, and if the server unable to respond immediately, it will send an empty ACK message. When the server is able to respond, it send a CON message with data.

3. Non Confirmable Request and Response:

Here a NON message is send by the client to the server, the server response with a NON message. No ACK is send.



Piggyback Message



Separate Message

TRANSPORT PROTOCOLS

Two important transport protocols for IoT infrastructure are,

1. BLE (Bluetooth Low Energy)
2. Li-Fi (Light Fidelity)



Bluetooth® Low Energy



BLE (Bluetooth Low Energy)

- It is a wireless protocol for connecting devices
- It is low power
- It is low cost
- Used for connecting IoT devices
- It uses 2.4GHz ISM band
- It is introduced in the Bluetooth 4.0 version.
- It is implemented in most of the smartphones.
- It is supported by most of the Operating Systems.

Architecture of BLE

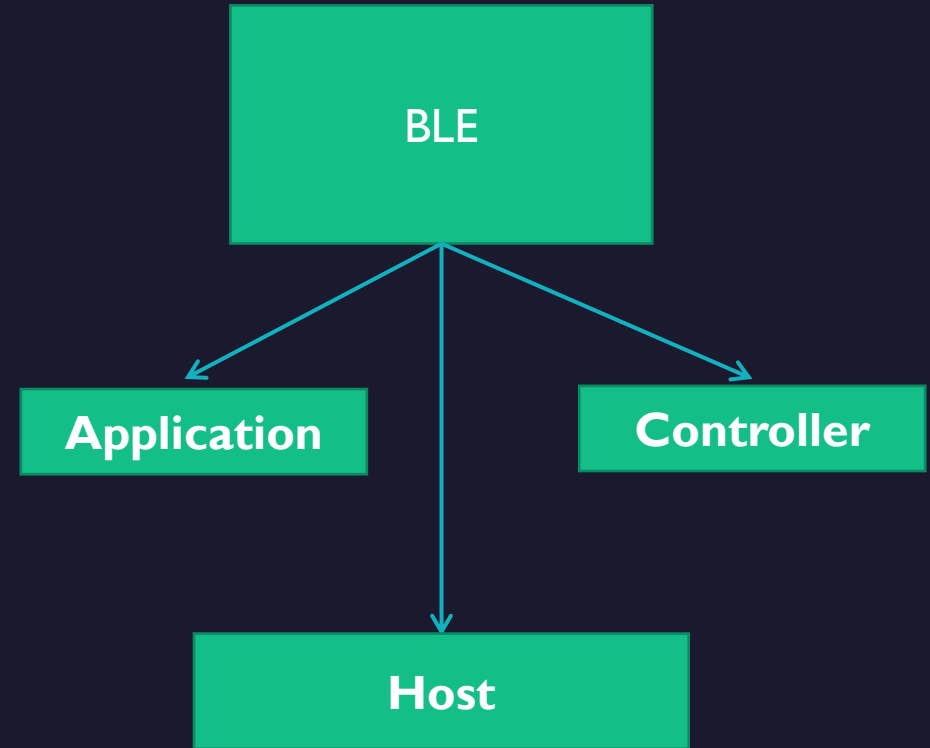
BLE has 3 components:

1. Application

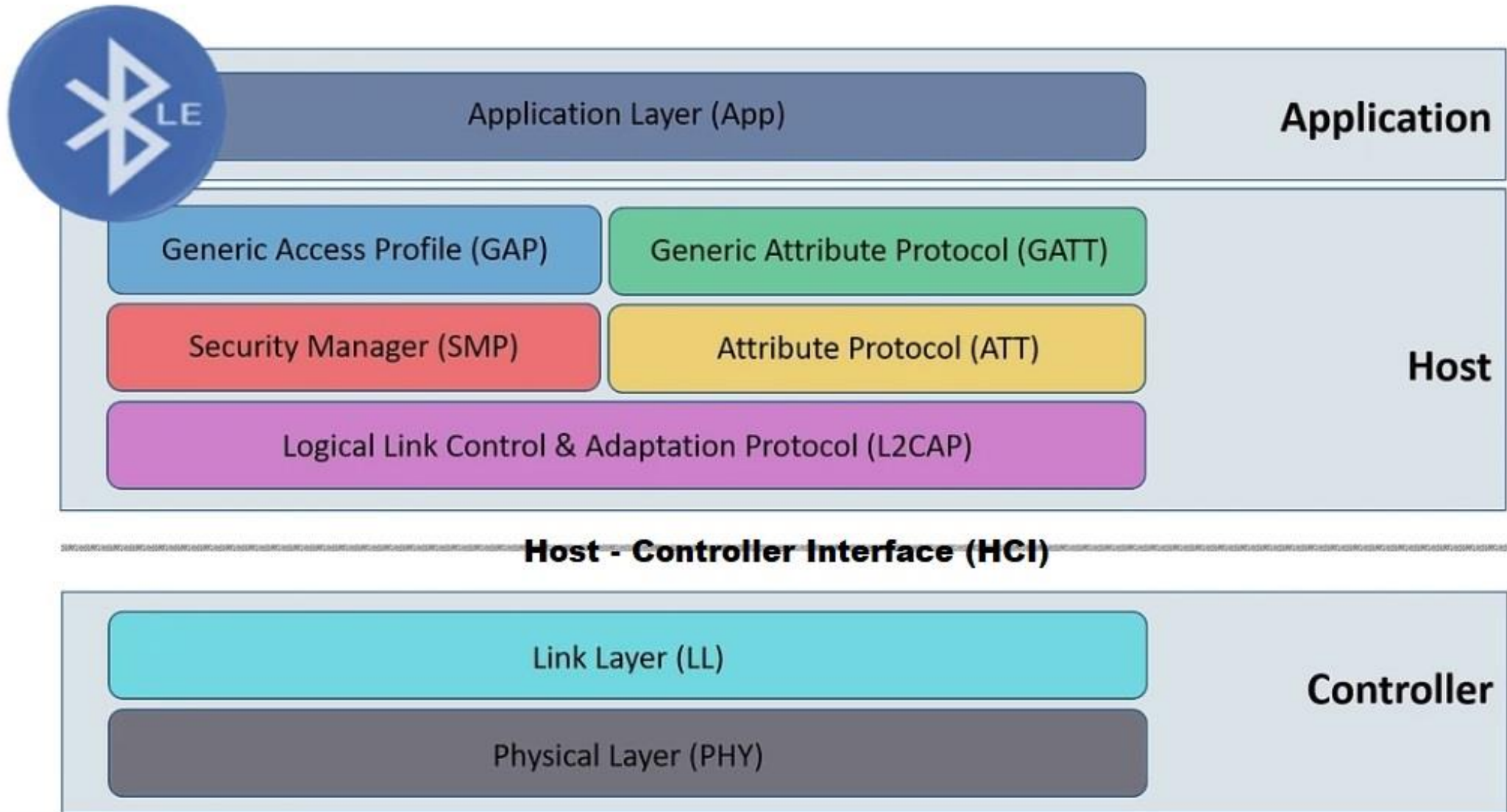
2. Host

3. Controller

- Application: It represent all applications.
- Host: It is responsible for connection establishment
- Controller: It takes care of data transmission and control



BLE Protocol Stack



BLE (Bluetooth Low Energy)



Controller :

Controller has two layers:

1. Link layer
2. Physical layer

Link Layer:

- It defines packet structure and control.

Physical Layer:

- It handles data transmission.
 - Modulation and demodulation.
 - Analog to digital and digital to analog conversion
- 

Host :

Host contains 5 layers:

1. Generic Access Profile (GAP)

- It handles device discovery, connection establishment and management.

2. Generic Attribute Profile (GATT)

- It defines the guidelines for data read/write .

3. Attribute Protocol (ATT)

- It defines the protocol for accessing data.

4. Logical Link Control and Adaptation Protocol (L2CAP)

- It is responsible for fragmentation and reassembly of data.
- It also do multiplexing and demultiplexing of channels.

5. Security Manager (SM)

- It manages pairing, authentication and encryption.



Application :

- Application layer is responsible for User interface (UI) and Application logic.

Host-Controller Interface :

- It act as an interface between Host and Control layers.
- It helps to connect devices from different manufactures.

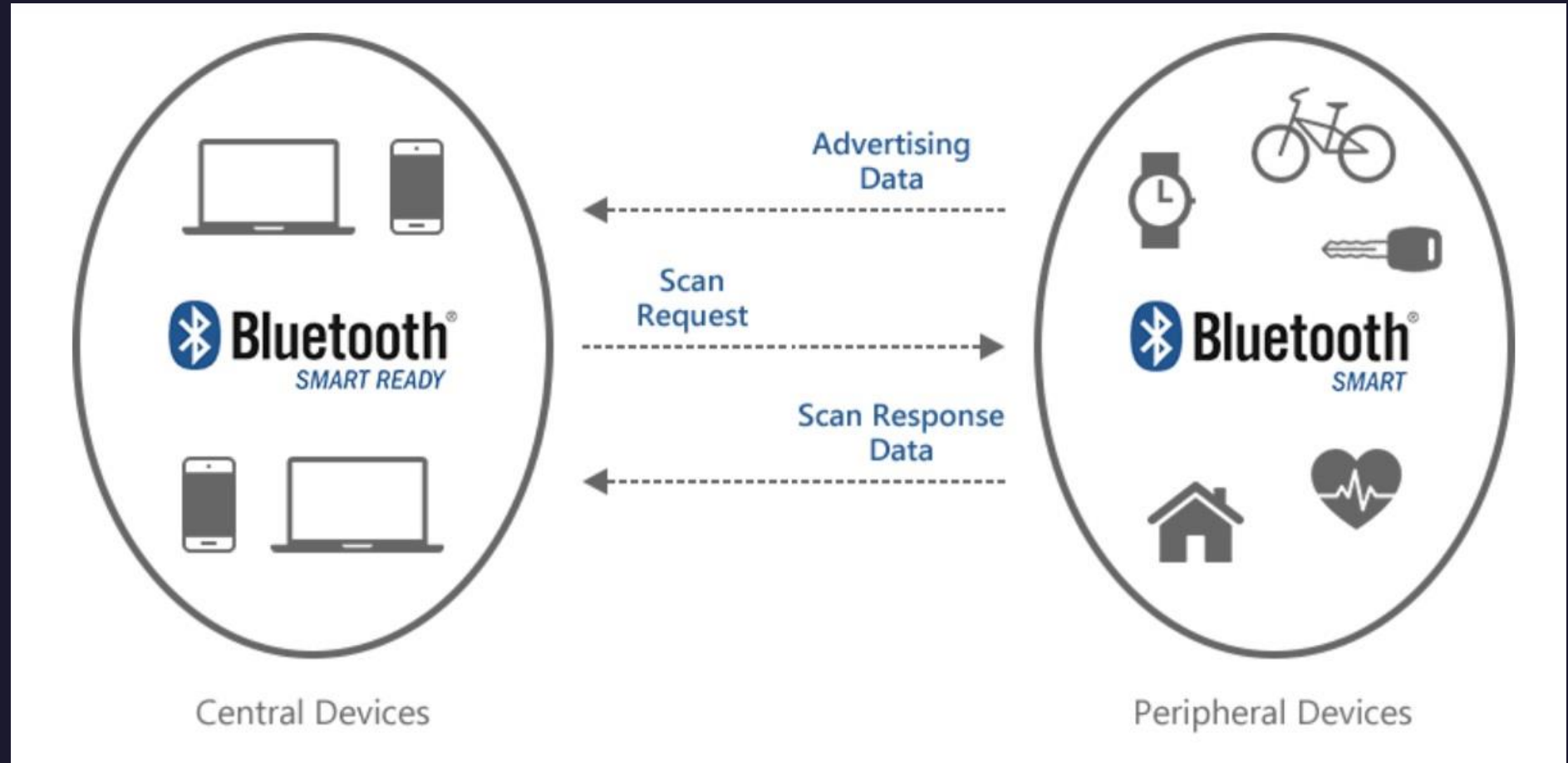


BLE Device connection

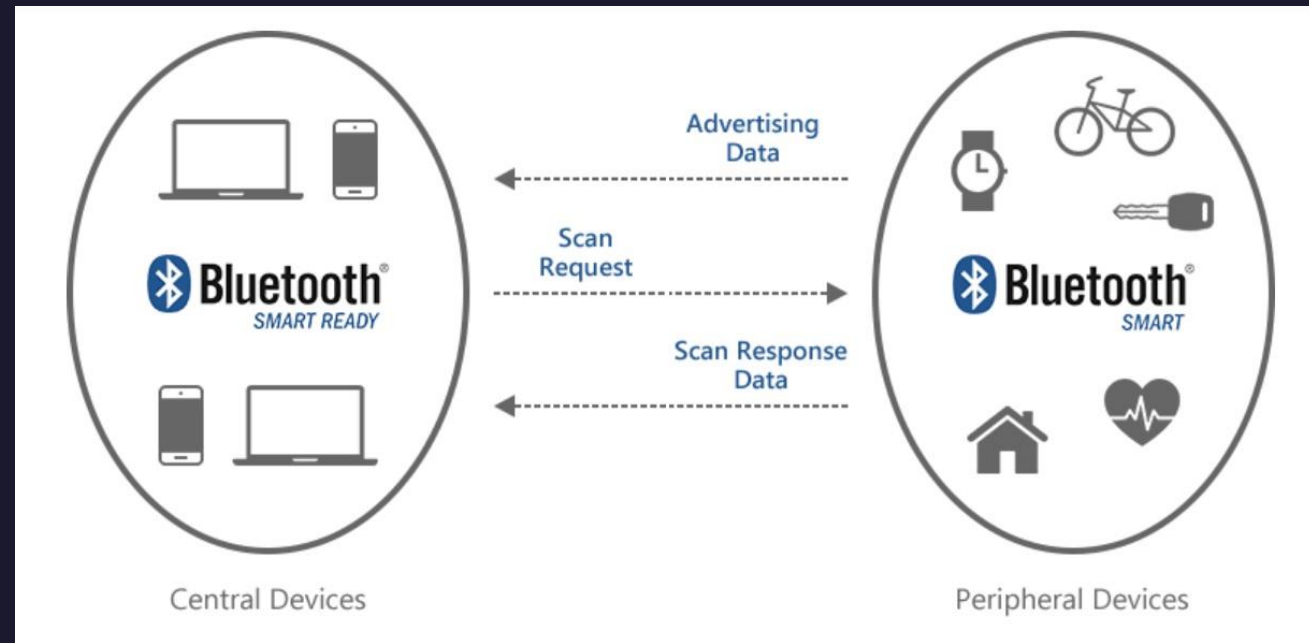
In a BLE network, two types of devices are there,

1. Peripheral

2. Central

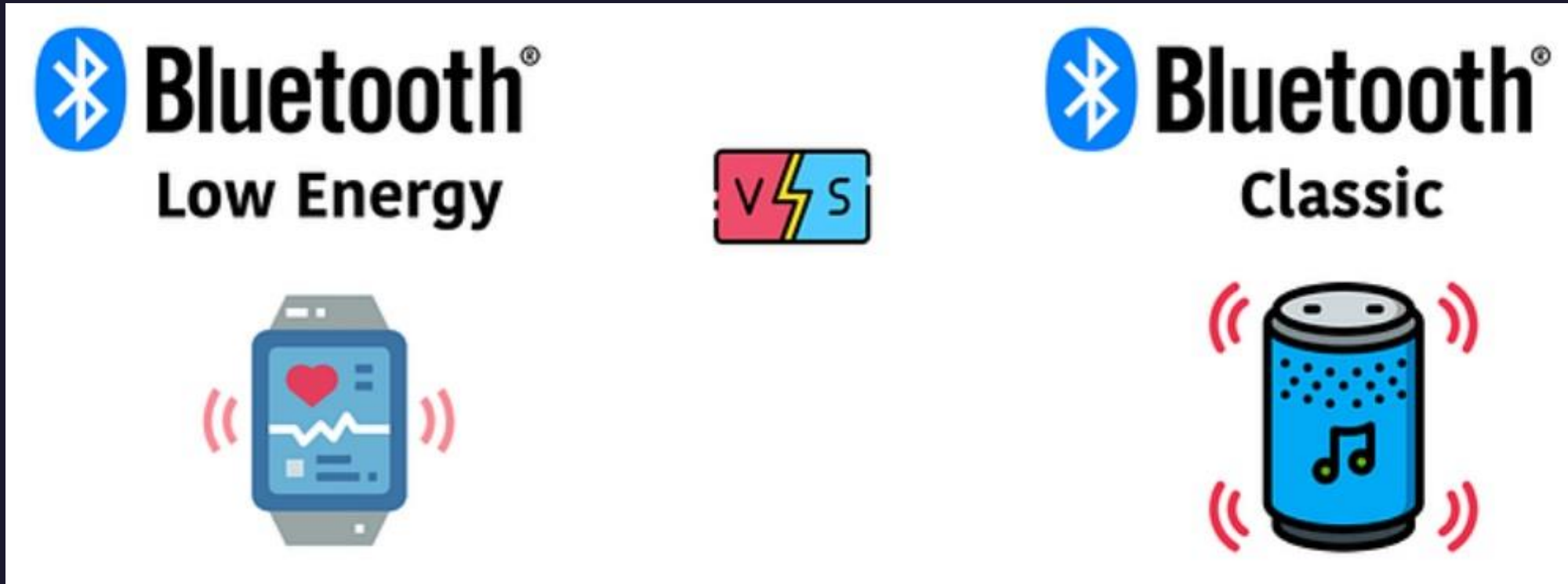


BLE Device connection



- The peripheral device broadcast advertising data.
- The central device scan for advertising data.
- If any advertising message is found, central device send scan request.
- On receiving the scan request, the peripheral device send response.
- Now connection is established and data can be send between central and peripheral devices.

Difference between BLE & Bluetooth Classic



	Bluetooth Classic	Bluetooth Low Energy
Usage	Used for audio streaming, file transfers, and headsets	Used for sensor data, control of devices, and low-bandwidth applications
Data Rate	Maximum data rate (2.1 Mbps)	The max data rate is at 1 Mbps
Power Consumption	High power consumption Radio is always on	Lower power consumption
	Shorter battery life	More efficient radio usage and on-time
		Longer battery life
RF Channel Spectrum	Operates over 79 RF channels	Operates over 40 RF channels
Range	Up to around 100 m	Lessthan 100 m
Topology	Peer-to-peer (1:1)	Peer-to-peer (1:1)
		Star topology (many:1)
		Broadcast (1:many)
		Mesh (many:many)
Smartphone Compatibility	Available on 100% of smartphones	Available on ~100% of smartphones
Cost of chipsets	Higher cost	Lower cost





TM



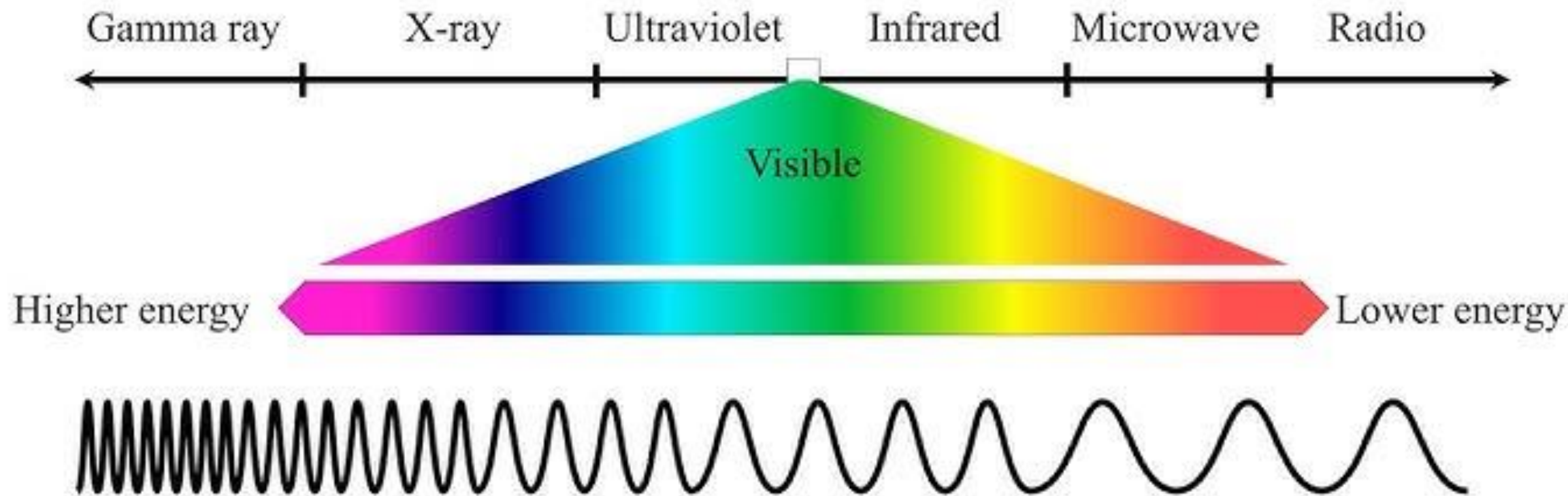
WI-FI

stands for

**Wireless
Fidelity**



Electromagnetic Spectrum



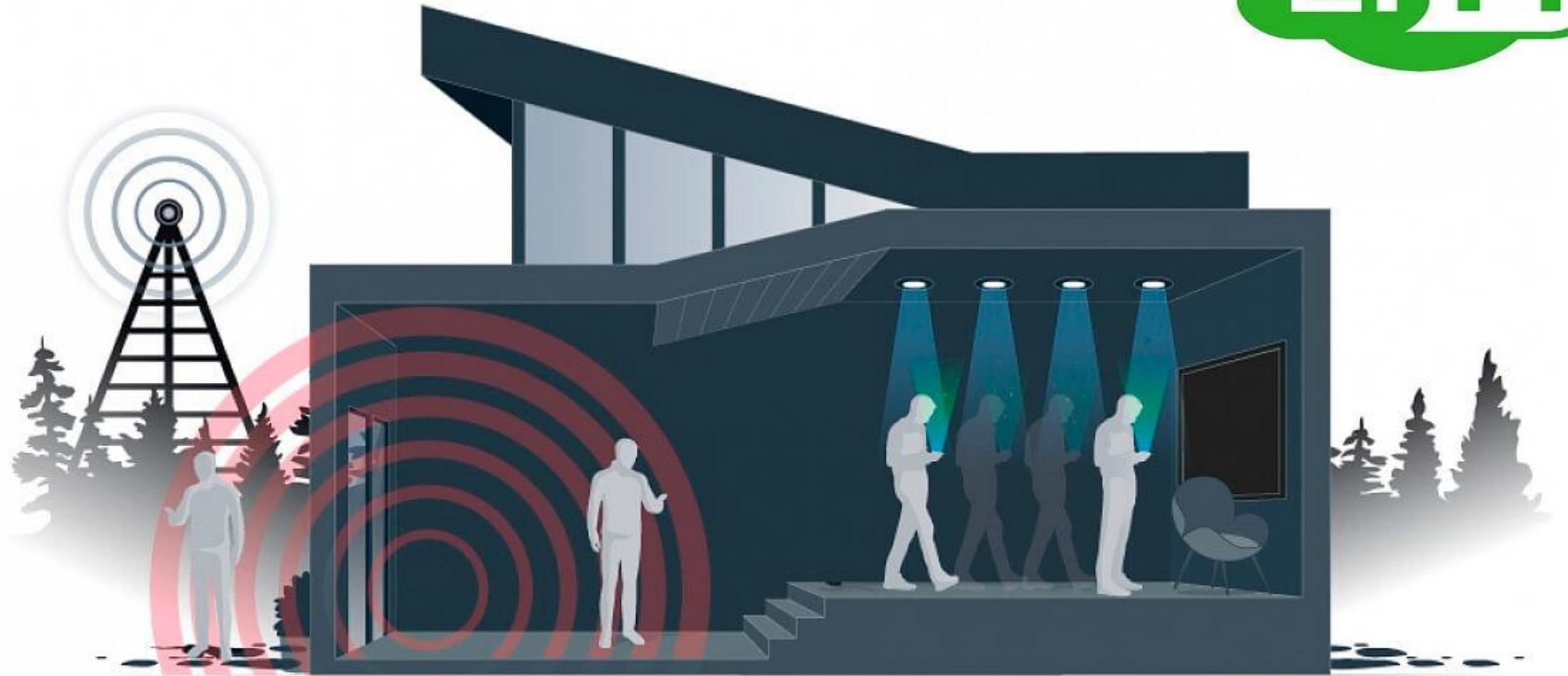


Radio Waves	→	Expensive, Less Secure
Infrared Rays	→	Causes Eye problem
Visible Rays	→	Best to Use
Ultra violet Rays	→	Effects Human Body
X-Rays	→	Health Issues
Gamma Rays	→	Dangerous

Li-Fi Technology



- Li-Fi means Light Fidelity
- Li-Fi is the transmission of data through light.
- Light source like LED bulb is used as transmitter.
- Change in the intensity of light is considered as binary zero and one.
- Changing the brightness occur very fast and human eye can not able to detect it.



Wi-Fi

LiFi



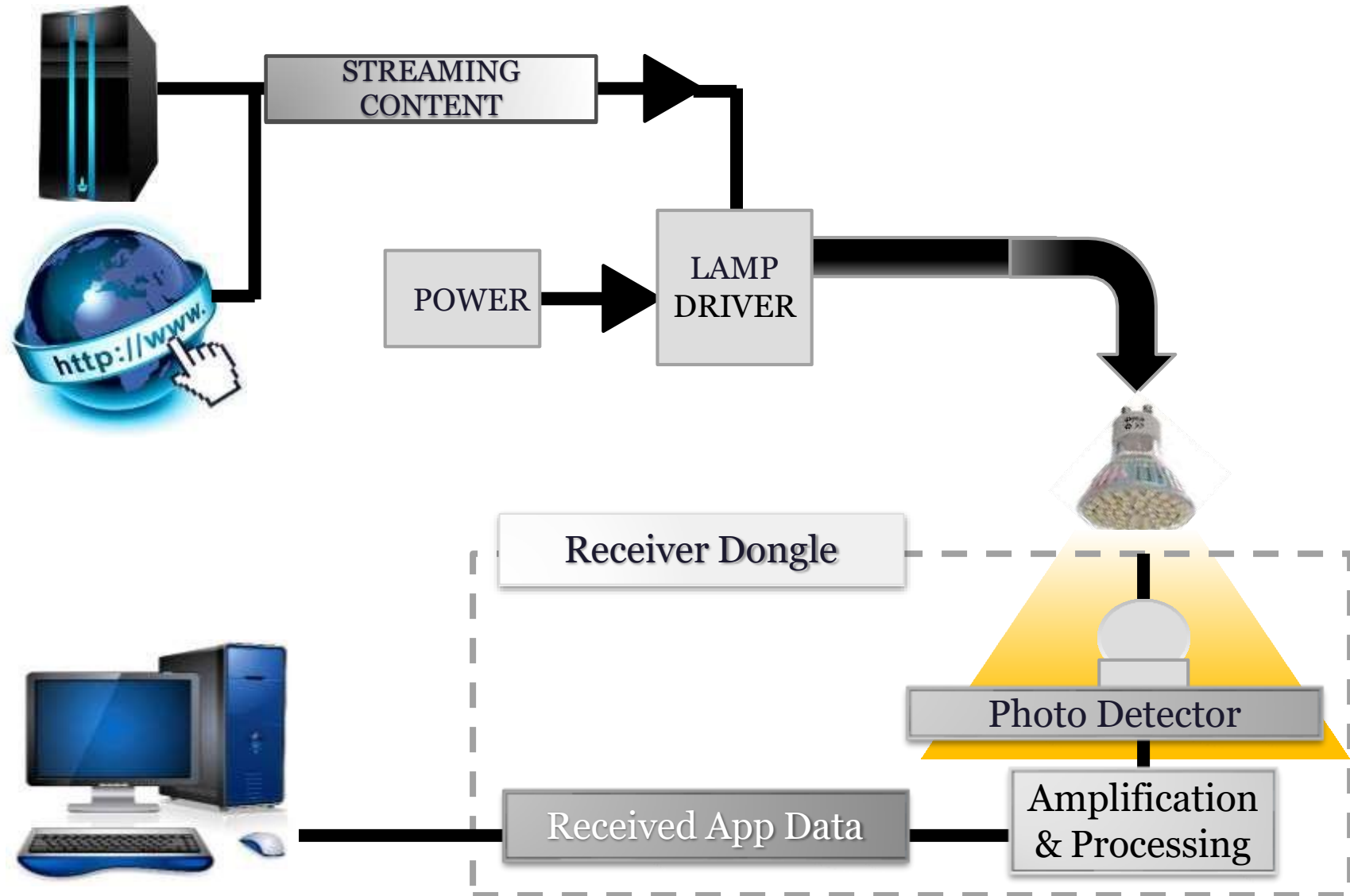
Li-Fi Technology



- Li-Fi was invented by a German Physicist
HARALD HAAS
in the year 2012



Li-Fi Working



Li-Fi Working

Transmitter:

- The streaming data is given to the LED lamp driver.
- The lamp driver changes the intensity of the lamp according to the binary value of the data.
- This happens very fast and human eyes can not really feel or see this.

Li-Fi Working

Receiver:

- The receiver contains a photo diode which detects the change in intensity of the light and generate electric current.
- After amplification and processing, the data is forwarded to the receiver.

Li-Fi Advantages

- Hight data transfer rate – 224Gbps
- Hight security
- Low power consumption
- Less harmful to humans
- High efficiency



Li-Fi Disadvantages

- Light can't pass through objects.
- Short range
- Interference from sunlight or external bulbs.
- Always requires light as medium.
- Challenges of how the receiving device will transmit back to transmitter



IPv4 – Internet Protocol - Version 4

- An IP address is a unique identification for a node connected to a network.
- Networks using TCP/IP uses this address to route messages.
- IP address has two versions – IPv4 and IPv6
- IPv4 address is 4 bytes or 32 bit long.
- It can be represented as binary or dotted decimal notation format.
- Example:
 - 172.16.0.1

IPv4 – Internet Protocol - Version 4

172.16.0.1

1st Octet

10101100

172

2nd Octet

00010000

16

3rd Octet

00000000

0

4th Octet

00000001

1

32 bits divided into 4 octets

IP Address Classes

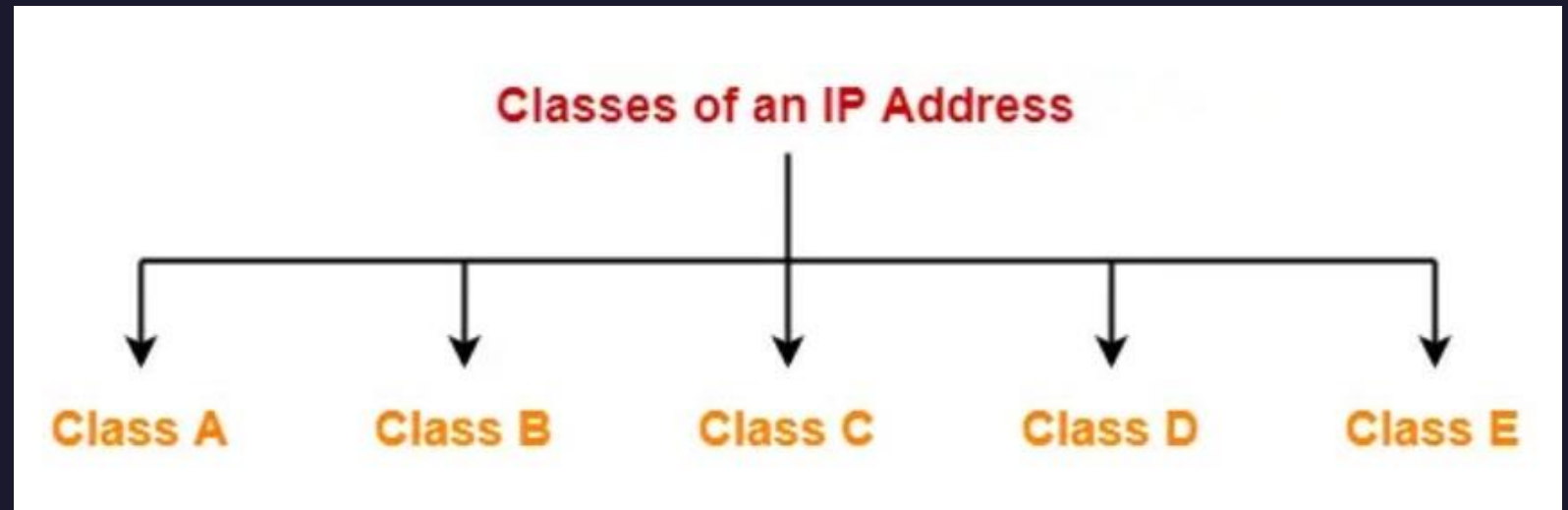
- An IP address has two sections:
 1. Net Id – Used to identify network.
 2. Host Id – Used to identify host.

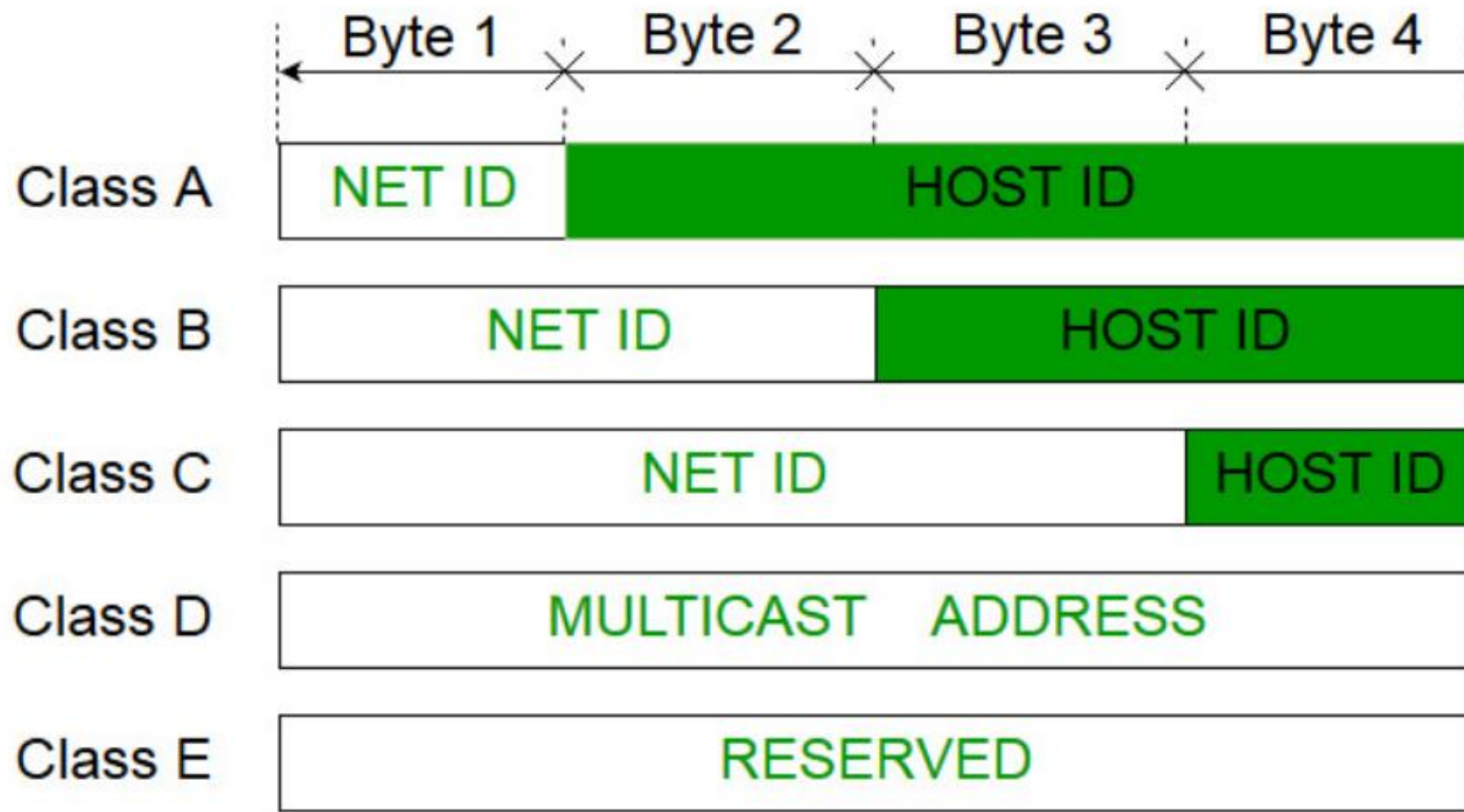


IP Address Classes

- IP address are classified into 5 classes:

1. Class A
2. Class B
3. Class C
4. Class D
5. Class E





Class A:	0	7 bits Network ID	24 bits Host ID	Very large Network (2^24 hosts !)		
Class B:	1	0	14 bits Network ID	16 bits Host ID	Medium size Network (Most popular !!!)	
Class C:	1	1	0	21 bits Network ID	8 bits Host ID	Small Network
Class D:	1	1	1	0	Multicast Group ID	Multicast Address
Class E:	1	1	1	1		Reserved (unused)

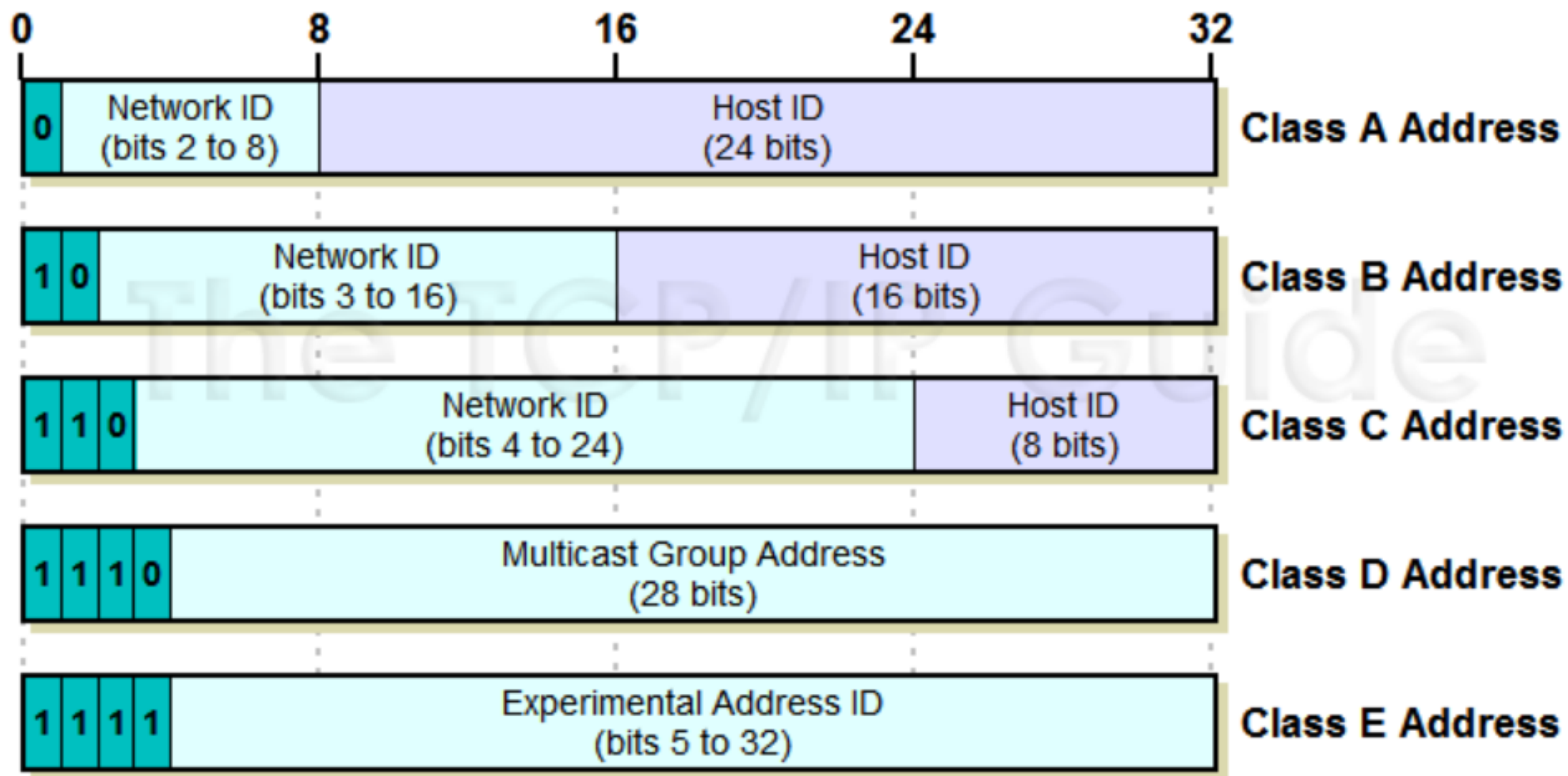


Figure 62: IP Address Class Bit Assignments and Network/Host ID Sizes

I. Class A (1.0.0.0 to 126.0.0.0):

- The first bit in a Class A address is always "0."
- These addresses were used for large networks, such as universities and major corporations.
- Class A addresses provide a very large number of host addresses (approximately 16 million) in each network.

2. Class B (128.0.0.0 to 191.255.0.0):

- The first two bits in a Class B address are "10."
- Class B addresses were typically used by medium-sized organizations and businesses.
- Each Class B network can have around 65,000 host addresses.

3. Class C (192.0.0.0 to 223.255.255.0):

- The first three bits in a Class C address are "110."
- Class C addresses were designed for smaller networks, such as small businesses.
- Each Class C network can support approximately 254 host addresses.



4. Class D (224.0.0.0 to 239.255.255.255):

- Class D addresses are reserved for multicast groups, which allow for one-to-many or many-to-many communication.
- These addresses are not assigned to individual hosts or networks.



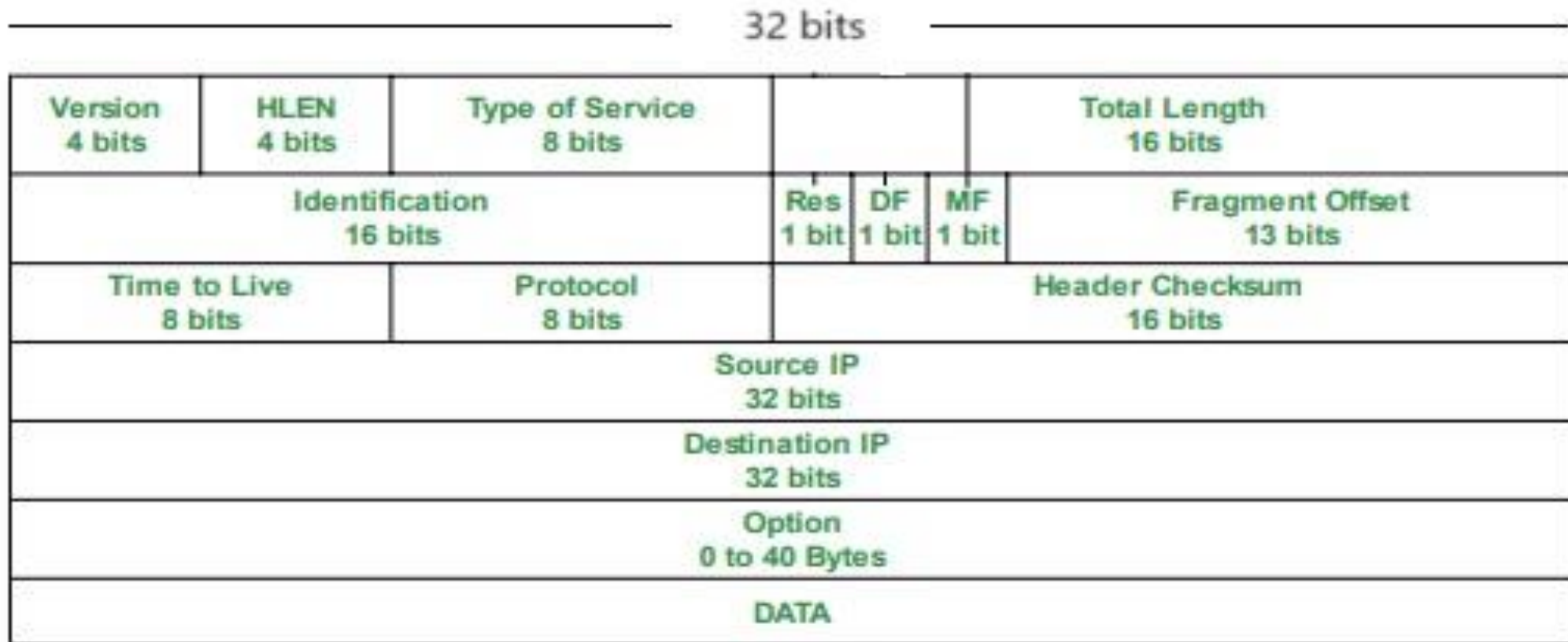
5. Class E (224.0.0.0 to 239.255.255.255):

- Class E addresses are reserved for experimental purposes and should not be used in regular networks.



Class	1st octet of IP address	Default Subnet Mask	Network / Host	Number of networks	Maximum nodes in a network
A	1 - 126	255.0.0.0	N.H.H.H	126	16,777,214
B	128 - 191	255.255.0.0	N.N.H.H	16,384	65,534
C	192 - 223	255.255.255.0	N.N.N.H	2,097,152	254
D	224 - 239				
E	240 - 254				

IPv4 Frame Format



Version(4 bits):

- It indicates the version number of IP.

Internet Header Length(4 bits):

- It indicates the length of the header in 32bit words.

Type of Service(8 bits):

- It indicates Quality of Service(QoS) such as delay, throughput, reliability etc.

Total length(16 bits):

- It indicates total length(header + data) of the IP packet.
- It is 20 bytes minimum and 65535 bytes maximum.

Identification(16 bits):

- It is used to identify packets of the same message.

Flags(3 bits):

- Res – Reserved
- DF - Don't fragment
- MF - More Fragments

Fragment offset(13 bits):

- When a message is split into different fragments, this offset is used to indicate the position of the fragment in the message.

Time to Live(8 bits):

- It indicates the life time of the packet in the network.
- Every time an IP packet passes through a router, TTL is decremented by 1.
- When it becomes 0, the packet is dropped by the router.

Protocol(8 bits):

- It indicates next higher level protocol

Header checksum(16 bits):

- It is used for header error detection.

Source IP address(32 bits):

- It indicates IP address of the source.

Destination IP address(32 bits):

- It indicates IP address of the destination.

Options:

- To set additional options.

Padding:

- These are the extra bits added to header to make header a multiple of 32bits..

IPv6– Internet Protocol - Version 6

- IPv6, is a network layer protocol used for communication over the Internet.
- It is the successor to IPv4.
- It was developed to overcome the address limitations of IPv4.

Features of IPv6

1. **Expanded address space:** IPv6 uses 128-bit addresses, which provides a larger address space.
2. **Address Format:** IPv6 addresses are represented as **eight groups of four hexadecimal digits** separated by colons
(2001:0db8:85a3:0000:0000:8a2e:0370:7334)
3. **Simplified Header:** The IPv6 header is simpler and more efficient than the IPv4 header.

Features of IPv6...

4. **Autoconfiguration:** IPv6 includes features for automatic address assignment and network configuration.
5. **Improved Security:** IPv6 includes built-in support for IPsec (Internet Protocol Security).
6. **Multicasting:** IPv6 has built-in support for multicast communication.

IPv6 Header Format

0	15	16	31
0 1 2 3 4 5 6 7	0 1 2 3 4 5 6 7	0 1 2 3 4 5 6 7	0 1 2 3 4 5 6 7
Version (4)	Traffic Class (8)	Flow label (20)	
Payload length (16)		Next header (8)	Hop limit (8)
Source Address (128)			
Destination Address (128)			

IPv6 Header format....

- **Version (4 bits):** This field identifies the version of the IP protocol.
- **Traffic Class (8 bits):** This field is used to prioritize or classify packets.
- **Flow Label (20 bits):** It is used by the source to label the packet belonging to same flow.
- **Payload Length (16 bits):** This field indicates the length of the IPv6 data portion.
- **Next Header (8 bits):** This field specifies the protocol that is immediately follows IPv6 protocol.

Features of IPv6

- **Hop Limit (8 bits):** This field limits the number of hops (routers) a packet can traverse before it is discarded.
- **Source IP Address (128 bits):** This field contains the IPv6 address of the packet's source.
- **Destination IP Address (128 bits):** This field contains the IPv6 address of the packet's intended destination.



URI – Uniform resource identifier

Definition:

- A Uniform Resource Identifier (URI) is a string of characters used to identify a resource on the internet, either by location, name, or both.

URI – Uniform resource identifier

Components:

A URI typically consists of two main components:

1. URL (Uniform Resource Locator):

Specifies the resource's location, including the protocol (e.g., HTTP, FTP), domain name or IP address, and the path to the resource (e.g., a webpage or file).

2. URN (Uniform Resource Name):

Identifies a resource by name but does not necessarily provide a means of locating it.

URI – Uniform resource identifier



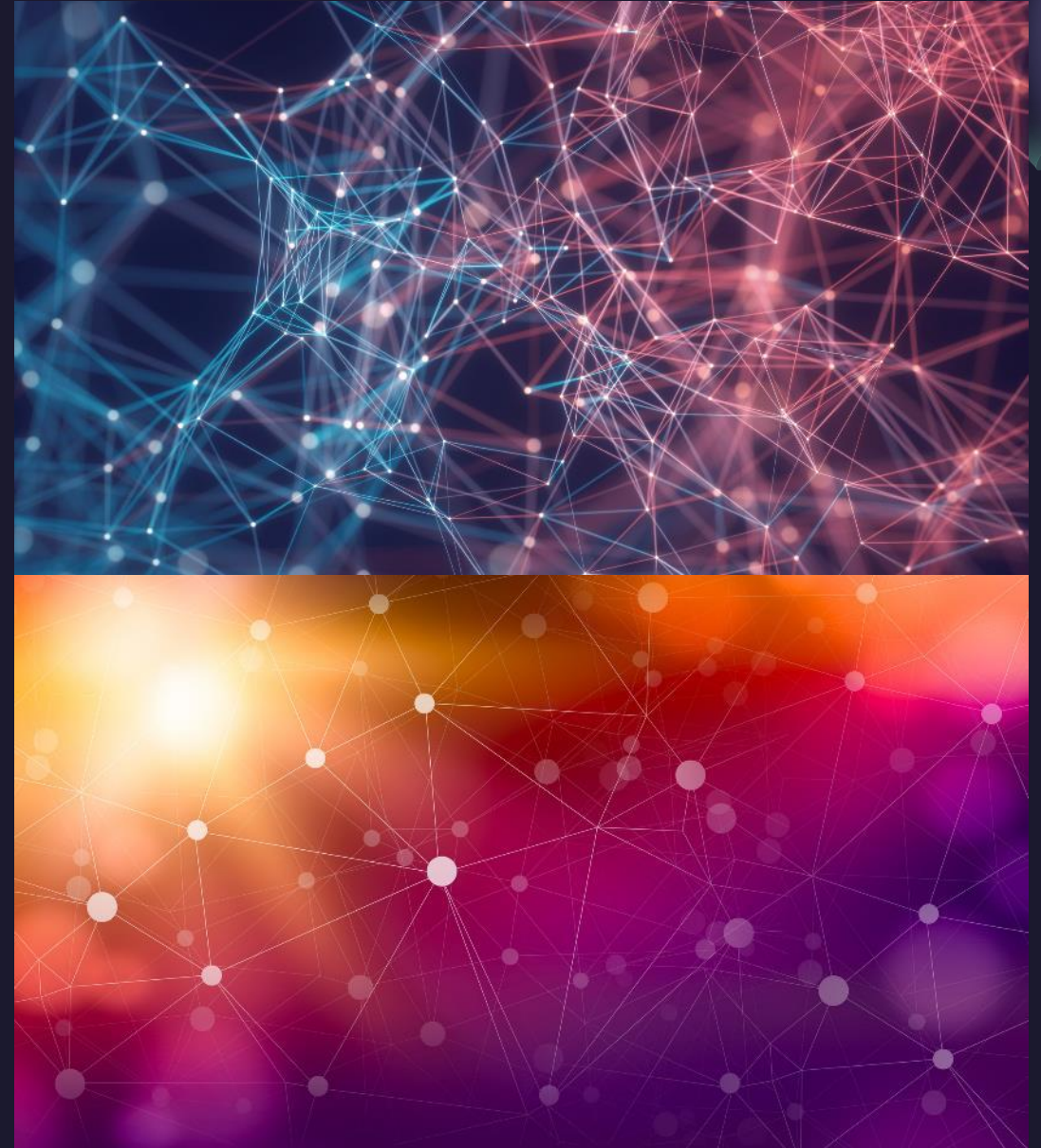
Examples:

URL: <https://www.example.com/index.html>

URN: urn:isbn:0451450523 (used to uniquely identify books)



Thank You





Summary